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Short note

Low-cost multi-modal microscope using Raspberry Pi

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ABSTRACT

This report describes a direction-dependent optical microscope operated with Raspberry Pi, with LED array illumination patterns controlled by Raspberry Pi. Images are obtained with a Raspberry Pi camera module. We demonstrate the use of Raspberry-Pi-based microscopes for multimodal imaging including bright-field microscopy, dark-field microscopy, differential phase-contrast microscopy, and polarization microscopy. We also demonstrate acquisition of direction-dependent dark-field and differential phase-contrast images. From a pair of opposite line or rectangular illumination, structures along the illumination axis were enhanced in dark-field images and differential phase-contrast images. Raspberry-Pi is open-source hardware. Raspberry-Pi-based compact, cost-effective, and multi-contrast microscopes are expected to be useful for science, technology, engineering and mathematics (STEM) education.

1. Introduction

Optical microscopy has been a useful tool for medical diagnosis, biological research, hospital examinations, education, and other fields, providing means to observe cellular structures and their dynamics. Because of microscopes' versatility and low price, wide-field optical microscopes with ambient light have been widely adopted for use in science, technology, engineering and mathematics (STEM) education. Enhanced visualization by smartphones, tablet computers, and compact cameras plays a role in STEM education with optical microscopes [1,2]. Different imaging modes such as bright-field, dark-field, phase-contrast, and polarization imaging provide complementary information for a single sample. Because each mode relies on a different optical hardware configuration, requiring inserts at the condenser aperture, polarization components, and/or specialized objectives, most mobile-phone-based optical microscopes produce bright-field images or fluorescence images [2–4]. Also, dual-mode mobile phone microscopy for bright-field and dark-field images has been reported [5]. Smartphone microscopes with LED arrays can provide simultaneous acquisition of bright-field, dark-field, and differential phase-contrast (DPC) by changing the LED array illumination pattern [6,7]. In fact, an optical microscopic system comprising two mobile phones has been demonstrated: one for illumination and the other for multi-modal imaging [8]. Unfortunately, despite providing the benefits explained above, cell phone-based systems are more expensive than other mini-microscopes. Operating programs and application software are difficult to produce.

Raspberry Pi is a microcomputer with onboard HDMI output, a USB port, a LAN port, and Wi-Fi. Moreover, an operating system can be installed on it; programming languages such as Python are available for it. Raspberry Pi was originally developed to promote

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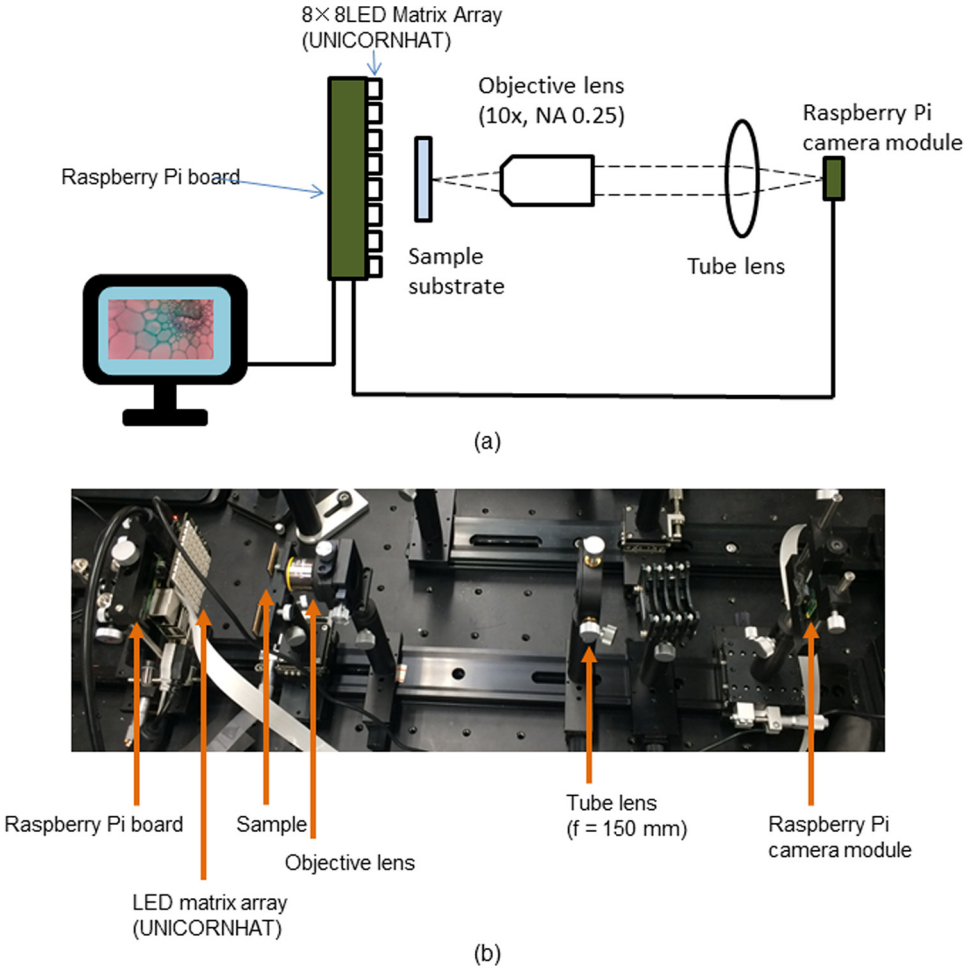

Fig. 1. a) Schematic for Raspberry pi optical microscope and (b) a photograph of the experimental setup.

the teaching of basic computer science in schools and in economically developing countries. Among other applications, Raspberry Pi technology has been used to construct cost-effective, miniaturized microscopes [9–14]. Several Raspberry-Pi-based microscopes have been reported: a bright-field microscope [9], a 3-D printed microscope [10,11], a time-lapse imaging microscope with an incubation system [12], a fluorescence microscope [13], and a field-portable multi-spectral microscope [14]. Microscopes with a controllable illumination source with an LED array provide a Fourier ptychography microscopy [15,16] and multi-contrast microscopy [17].

This report describes a low-cost multimodal microscope with an LED array operated with Raspberry Pi. The LED array illumination patterns are controlled using Raspberry Pi. Various images are obtained with a Raspberry Pi camera module. We demonstrate the use of a Raspberry-Pi-based microscope for multimodal imaging, including bright-field microscopy, dark-field microscopy, differential phase-contrast microscopy, and polarization microscopy. We also demonstrate acquisition of direction-dependent dark-field and differential phase-contrast images. From a pair of opposite line or rectangular illumination, structures along the illumination axis were enhanced in dark-field images and differential phase-contrast images. Raspberry-Pi is open-source hardware. Raspberry-Pi-based compact, cost-effective, and multi-contrast microscopes are anticipated for wider use in education and remote cell monitoring.

2. Materials and Methods

A schematic for a Raspberry-Pi-based LED array microscope was presented in an earlier report [17]. Fig. 1(a) and (b) respectively portray a schematic and a picture of the experimental system. The microscope has a 10× objective lens (LMPlanFL 10×; Olympus Corp.) with a numerical aperture (NA) of 0.25, a 150 mm focal length tube lens (DLB-30-150PM; Sigma Koki Co. Ltd.), and a Raspberry Pi camera module (UD-RPCAMERA). The Raspberry Pi camera module captures images of 3280 × 2464 pixels, each of 1.12 μm × 1.12 μm. An RGB LED array (UNICORNHAT; Pimoroni Ltd.) was used as the light source. It has a matrix of 64 (8 × 8) RGB LEDs. The LED array total size was 53 mm × 53 mm. The pitch of each LED was 5 mm. The illumination patterns of the LED array were controlled by the Raspberry Pi. Images were obtained using the Raspberry Pi camera module. Images were captured in JPEG format. After capturing images, 8-bit RGB images were converted into gray scale. Programming code was written in Python.

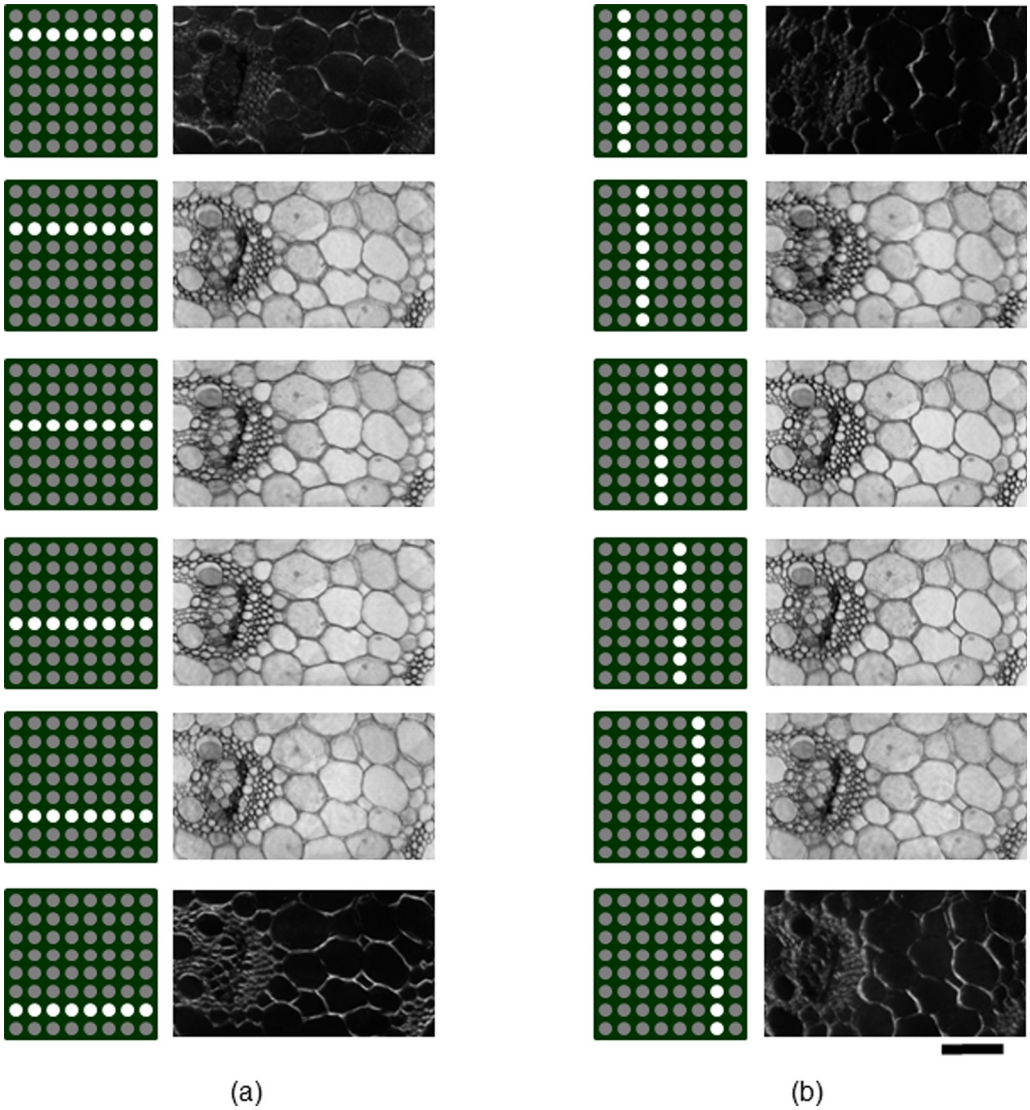

Fig. 2. Relation between lighting positions and images acquired using line-shape illumination with the LED array. The sample was a cross-section of a corn stem. Scale bar: 100 μm. Left panels present line illumination patterns of an LED array: (a) horizontal direction and (b) vertical direction.

The distance between the sample and the LED array was set as 5 cm. As samples, we used prepared microscope slides.

3. Results and Discussion

3.1. Imaging with line illumination

First, we investigated acquired images with line-shape oblique illumination. Schematics for line illumination patterns are presented with acquired images in Fig. 2. Eight LEDs in a row were lit up sequentially in the horizontal direction from 1 to 8, as shown in the left panel. As presented in the left panel of Fig. 2, the LED array was lit up in a row sequentially in the horizontal direction from the second to seventh row. The position between rows 4 and 5 in the LED array was set to be the optical axis. At the second-row and seventh-row illuminations, dark-field images were obtained. At the third-row to sixth-row illuminations, bright-field images were obtained. Then, 8 LEDs in a column were lit up sequentially toward the vertical direction, as shown from 1 to 8 in right panel in Fig. 2. At second-column and seventh-column illumination, dark-field images were obtained. At the third-column to sixth-column illumination, bright-field images were obtained. In dark-field images with vertical and horizontal illuminations, vertical and horizontal cell walls were highlighted. The cell wall direction was visualized with the asymmetric illumination.

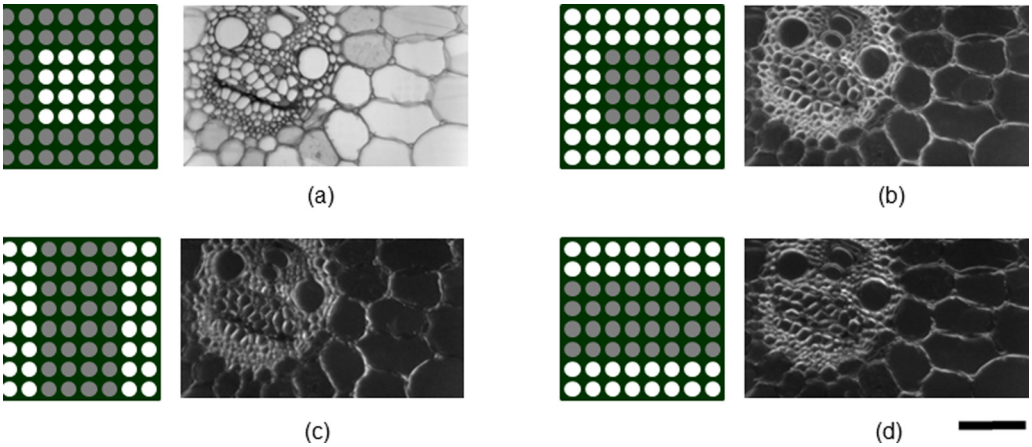

Fig. 3. Bright-field and dark-field images of a corn section taken by control of the LED array illumination patterns: (a) bright-field, (b) dark-field, (c) left–right dark-field, and (d) top–bottom dark-field images. The sample was a cross section of a corn stem. Scale bar: 100 μm.

3.2. Directional dependence of dark-field images with two-dimensional illumination

An LED array microscope can provide almost simultaneous acquisition of bright-field and dark-field images by changing the illumination patterns of the light source with no mechanical actuation [6–7]. We investigated the relation between two-dimensional illumination patterns and acquired images in this setup (Fig. 3). In experiments used to examine the bright-field light pattern (Fig. 3(a)), a square of central 4 × 4 LED arrays was illuminated. A dark-field image can be acquired simply by turning on the LEDs at the edge of the array (Fig. 3(b)). Using these setups, we obtained bright-field and dark-field images. Fig. 3(c) and (d) respectively show illumination patterns for the left–right dark-field and top–bottom dark-field. In dark-field images, different illumination patterns enhanced visualization of structures along the illumination axis [18].

Fig. 4(a) shows a bright-field image of a longitudinal section of the corn stem; Fig. 4(b) shows the dark-field image. Fig. 4(c) and (d) respectively depict the left–right and top–bottom dark-field images. In dark-field images, different illumination patterns enhanced visualization of structures along the illumination axis when the sample has structures along the illumination direction.

3.3. Directional dependence of differential phase-contrast images with two-dimensional illumination

The DPC images are calculated from differences of two images obtained with complementary rectangular illumination. Here, images obtained with the left rectangular and the right rectangular patterns are denoted, respectively, as I_L and I_R. Then, the left–right DPC image, I_{DPC_LR} is definable as [6,7]:

$$I_{DPC_LR} = \frac{I_L - I_R}{I_L + I_R} \tag{1}$$

Similarly, the top–bottom DPC image, I_{DPC_TB} is definable as

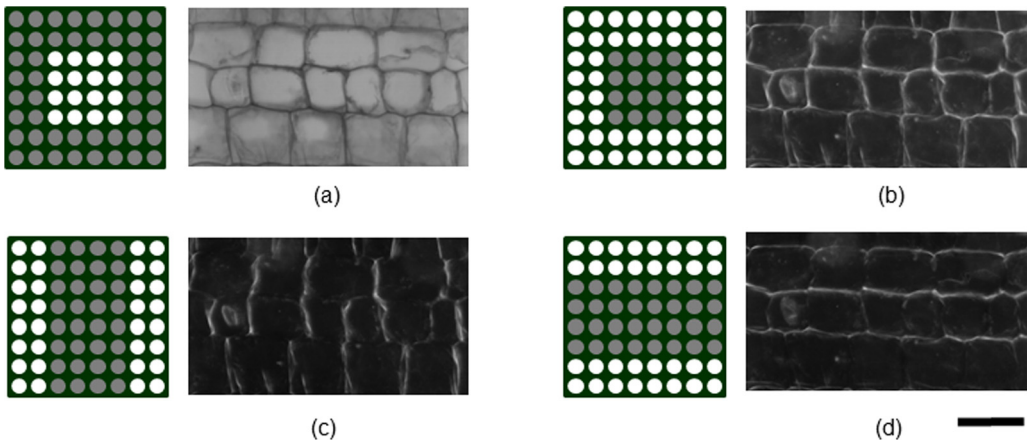

Fig. 4. Bright-field and dark-field images taken by control of illumination patterns of the LED array: (a) bright-field, (b) dark-field, (c) left–right dark-field, and (d) top–bottom dark-field images. The sample was a longitudinal section of a corn stem. Scale bar: 100 μm.

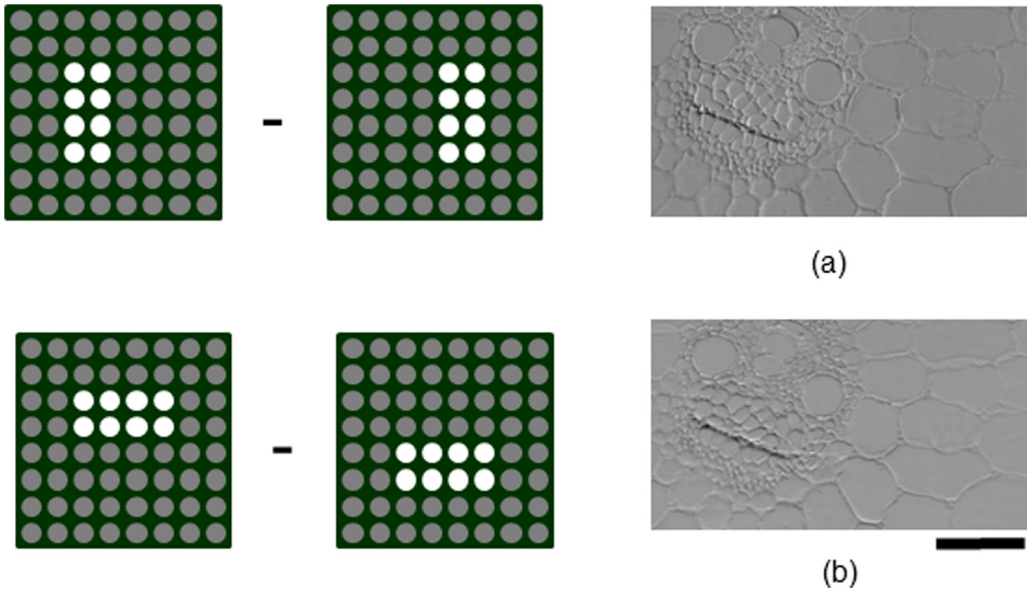

Fig. 5. Differential phase-contrast images of a corn section taken by control of illumination patterns of the LED array: (a) Left–right DPC image and (b) top–bottom DPC image. The sample was a cross section of the corn stem. Scale bar: 100 μm.

$$I_{DPC_TB} = \frac{I_T - I_B}{I_T + I_B}, \tag{2}$$

where I_T and I_B respectively denote images obtained with the top rectangle and bottom rectangle illumination. Computational imaging using software is necessary (Image J) to acquire DPC images using Eqs. (1) and (2).

Fig. 5(a) and (b) respectively show left–right and top–bottom DPC images of a corn stem cross-section. Fig. 6(a) and (b) respectively show left–right and top–bottom DPC images of longitudinal sections of a corm stem. By vertical or horizontal illumination, structures along the illumination axis were enhanced in DPC images.

3.4. Polarization microscope

A polarization microscope is an application of one possible contrast enhancement technique. Contrast can be enhanced by

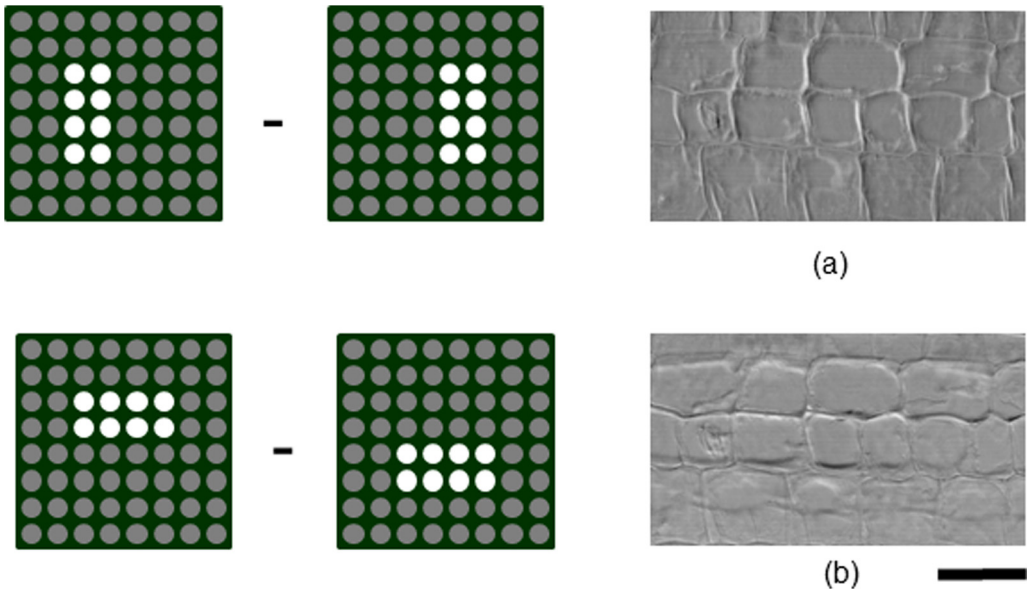

Fig. 6. Differential phase-contrast images of a corn section taken by control of illumination patterns of the LED array: (a) Left–right DPC image and (b) top–bottom DPC image. The sample was a longitudinal section of a corn stem.

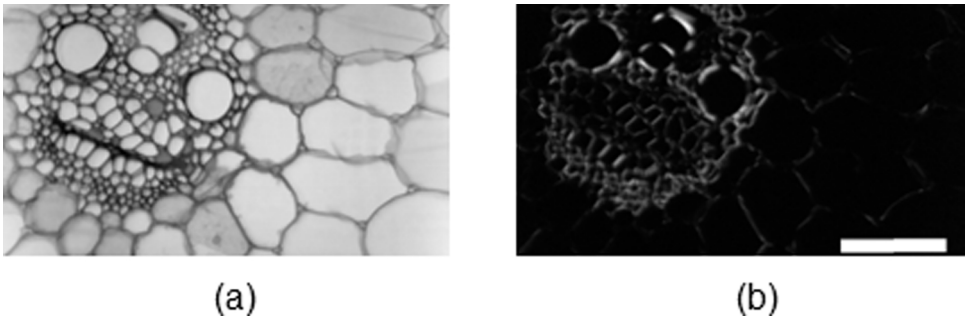

Fig. 7. Birefringent properties of cellular structures: (a) unpolarized transmission microscope image and (b) cross Nicole image of a corn stem. Scale bar: 100 μm.

birefringence properties of the sample. Here, we implemented a polarizational function in this Raspberry-Pi-based microscope. Polarizational dependence was investigated by setting two polarizers (#16179; Edmond Optics Inc.): simple addition of a polarizer after an LED array and an analyzer after the objective lens. Fig. 7(a) and (b) respectively portray the unpolarized transmission image and cross Nicole image of the sample. Results show structures of birefringence properties primarily by the cell wall of the vasculature.

The resolution of the constructed Raspberry-Pi microscope was measured. Bright-field images of positive and negative 1951 USAF test targets (#38257; Edmund Optics Inc.) are shown in Fig. 8(a) and (b). The contrast transfer function calculated from the images was greater than 0.95. The resolution was calculated as less than 4.4 μm.

3.5. Discussion

A salient benefit of Raspberry Pi microscopes is that it is possible to remove a lens from the camera module. As described later, in smartphone microscopes, aberrations and image magnification present challenges that hinder high-quality imaging. In our setup, the Raspberry Pi camera lens was removed. An optical system using an objective lens and a tube lens was adopted. This optical system presents important benefits in that the magnification is easily variable by changing objective lenses with different magnifications and that the aberration is small. However, a difficulty exists by which the cost of optical components such as an objective lens is high. The microscopic system is large.

In a smartphone microscope, a camera lens in the smartphone is a built-in type. To realize a microscope with a smartphone, it is common to use a ball lens [2,3], an inverted lens [5], or a conventional microscope objective lens. When a ball lens is used, only the central part of the image is clear, whereas the remainder of the area is blurred because of the effect of field curvature and spherical aberration [2,3]. Magnification, which is determined by the ratio of the focal lengths of the embedded camera lens and the ball lens, is small. Switz et al. developed an inverse dual lens system to reduce aberration [19]. The reverse lens can reduce aberration by adding the same type of lens in the opposite direction. Because the camera lens has the same focal length, the magnification is one. Changing the magnification is difficult. Therefore, in our setup, an optical microscopic system using an objective lens and a tube lens was adopted. Then an image sensor was placed by removing the Raspberry Pi camera lens.

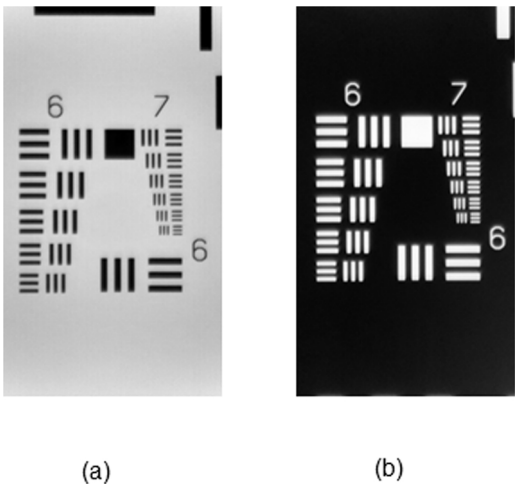

Fig. 8. Imaging of test target: (a) positive and (b) negative.

4. Conclusion and outlook

This report describes a multi-modal microscope operated with Raspberry Pi. Illumination patterns in the LED array were controlled with Raspberry Pi. Also, images were obtained using a Raspberry Pi camera module. The Raspberry-Pi-based LED array microscope offers functions of direction-dependence of the specimen. Bright-field and dark-field images were captured directly using the camera module. Line-shape illumination by line LEDs enables the acquisition of dark-field and bright-field images by changing its position. We demonstrated the acquisition of bright-field, dark-field, and differential phase-contrast images of cells. By vertical or horizontal illumination, structures along the illumination axis were enhanced in dark-field images and differential phase-contrast images. With the availability of cost-effective Raspberry-Pi systems, the reported technique is expected to foster opportunities for education and remote cell monitoring. Raspberry Pi facilitates connection to devices via Wi-Fi. This microscope can therefore transmit images to a host computer for remote monitoring. Taken together, compact, inexpensive raspberry-Pi microscopes are expected to be useful for medical diagnosis, water inspection, field surveys, and education applications.

Declaration of Competing Interest

None declared.

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References

[1] T. Hergemöller, D. Laumann, Smartphone magnification attachment: microscope or magnifying glass, The Physics Teacher 55 (2017) 361.
[2] Z.J. Smith, K. Chu, A.R. Espenson, M. Rahimzadeh, A. Gryshuk, M. Molinaro, D.M. Dwyre, S. Lane, D. Matthews, S. Wachsmann-Hogiu, Cell-phone-based platform for biomedical device development and education applications, PLoS One 6 (3) (2011) e17150.
[3] A. Arpa, G. Wetzstein, D. Lanman, R. Raskar, Single lens off-chip cellphone microscopy, IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), July, 2012.
[4] C. Fuhong, T. Wang, W. Lu, X. Zhang, High-resolution mobile bio-microscope with smartphone telephoto camera lens, Optik 207 (2020) 164449.
[5] A. Orth, E.R. Wilson, J.G. Thompson, B.C. Gibs, A dual-mode mobile phone microscope using the onboard camera flash and ambient light, Sci. Rep. 8 (2018) 3298.
[6] Z.F. Phillips, M.V. D'Ambrosio, L. Tian, J.J. Rulison, H.S. Patel, N. Sadras, A.V. Gande, N.A. Switz, D.A. Fletcher, L. Waller, Multi-contrast imaging and digital refocusing on a mobile microscope with a domed LED array, PLoS One 10 (5) (2015) e0124938.
[7] D. Jung, J.-H. Choi, S. Kim, S. Ryu, W. Lee, J.-S. Lee, C. Joo, Smartphone-based multi-contrast microscope using color-multiplexed illumination, Sci. Rep. 7 (2017) 7564.
[8] S. Kheireddine, A.S. Perumal, Z.J. Smith, D.V. Nicolau, S. Wachsmann-Hogiu, Dual-phone illumination-imaging system for high resolution and large field of view multi-modal microscopy, Lab Chip 9 (5) (2019) 825–836.
[9] K. Kaur, J. Singh, A Review for digital microscope using Raspberry Pi, International Journal on Recent and Innovation Trends in Computing and Communication 5 (1) (2017) 131–132.
[10] T. Baden, A.M. Chagas, G. Gage, T. Marzullo, L.L. Prieto-Godino, T. Euler, Open labware: 3-D printing your own lab equipment, PLoS Biol. 13 (5) (2015) e1002175.
[11] A.M. Chagas, L.L. Prieto-Godino, A.B. Arrenberg, T. Baden, The €100 lab: A 3D-printable open-source platform for fluorescence microscopy, optogenetics, and accurate temperature control during behaviour of zebrafish, Drosophila, and *Caenorhabditis elegans*, PLoS Biol 15 (7) (2017) e2002702.
[12] Z. Wang, A. Boddeda, B. Parker, R. Samanipour, S. Ghosh, F. Menard, K. Kim, A high-resolution mini-microscope system for wireless real-time monitoring, IEEE Trans Biomed Eng. 65 (7) (2018) 1524–1531.
[13] I. Nuñez, T. Matute, R. Herrera, R. Herrera, J. Keymer, T. Marzullo, T. Rudge, F. Federici, Low cost and open source multi-fluorescence imaging system for teaching and research in biology and bioengineering, PLoS One 12 (11) (2017) e0187163.
[14] J.L. Deglint, K. Schoneveld, F. Kazemzadeh, A. Wong, A compact field-portable computational multispectral microscope using integrated Raspberry Pi, Journal of Computational Vision and Imaging Systems 2 (1) (2016).
[15] P.C. Konda, T. Aidukas, A.R. Harvey, Miniature Fourier ptychography microscope using Raspberry Pi camera and hardware, OSA Technical Digest (online) (Optical Society of America), (2017) paper JTu5A.17.
[16] T. Aidukas, R. Eckert, A.R. Harvey, L. Waller, P.C. Konda, Low-cost, sub-micron resolution, wide-field computational microscopy using opensource hardware, Sci. Rep. 9 (2019) 7457.
[17] R. Maruyama, T. Shimada, R. Yokoe, R. Yokoe, H. Arimoto, W. Watanabe, Implementation of a Raspberry-Pi-based LED array microscope for multi-contrast images, Proceedings of SPIE 10711, in Biomedical Imaging and Sensing Conference (2018) 107111Y.
[18] R. Sugimoto, R. Maruyama, Y. Tamada, H. Arimoto, W. Watanabe, Contrast enhancement by oblique illumination microscopy with an LED array, Optik – International Journal for Light and Electron Optics 183 (2019) 92–98.
[19] N.A. Switz, M.V. D'Ambrosio, D.A. Fletcher, Low-cost mobile phone microscopy with a reversed mobile phone camera lens, PLoS One 9 (5) (2014) e95330.